Hereditary Hemorrhagic Telangiectasia Type 2 (HHT2): Comprehensive Disease Characteristics Report

Disease: Hereditary Hemorrhagic Telangiectasia Type 2 **MONDO ID:** MONDO:0010880 **OMIM:** #600376 (phenotype); *601284 (ACVRL1 gene) Orphanet: ORPHA:774 (parent HHT) | ICD-10: I78.0 | MeSH: D013683 Category:** Mendelian (autosomal dominant)

Summary

Hereditary Hemorrhagic Telangiectasia Type 2 (HHT2, Rendu–Osler–Weber syndrome, type 2) is an autosomal dominant vascular dysplasia caused by germline heterozygous **loss-of-function mutations in *ACVRL1*** (also called *ALK1*, on chromosome 12q13), which encodes the endothelial type I TGF-β/BMP receptor **activin receptor-like kinase 1 (ALK1)**. ALK1 binds the circulating ligands **BMP9 and BMP10** with high affinity and signals through **SMAD1/5/8** to maintain vascular quiescence and arterial identity. Haploinsufficiency of ALK1 de-represses angiogenesis, producing the two clinical hallmarks of HHT: fragile mucocutaneous **telangiectases** and larger visceral **arteriovenous malformations (AVMs)** that shunt blood directly from arteries to veins, bypassing capillary beds (PMID: 33513792, PMID: 16470787).

HHT2 has a **distinct genotype–phenotype profile** compared with *ENG*-associated HHT1: epistaxis begins later and is less penetrant, pulmonary and cerebral AVMs are fewer, but hepatic AVMs, gastrointestinal bleeding, and pulmonary arterial hypertension (PAH, ~20% of *ACVRL1* carriers) are more common (PMID: 17224686, PMID: 16470787, PMID: 42516693). The combined clinical prevalence of HHT is roughly **1 in 5,000**, though genomic-database analyses suggest the true genetic prevalence is 1.5–4-fold higher, indicating substantial underdiagnosis (PMID: 41610956).

Diagnosis combines the four clinical **Curaçao criteria** with molecular confirmation by *ACVRL1* sequencing, and management follows the **Second International HHT Guidelines** (2020): AVM screening and embolization, hematologic support for anemia, and disease-modifying **antiangiogenic therapies** (bevacizumab, pazopanib, thalidomide, and—newly

validated in a randomized trial—**pomalidomide**) (PMID: 32894695, PMID: 35226946, PMID: 39292928). Notably, *ACVRL1*/HHT2 genotype trends toward a **less robust pomalidomide response**, a first hint of genotype-guided therapy in HHT (PMID: 41719457).

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1. Disease Information

Overview. HHT2 is one subtype of hereditary hemorrhagic telangiectasia, a systemic autosomal dominant fibrovascular dysplasia characterized by the absence of normal capillary beds between arterioles and venules, leading to direct artery-to-vein connections. Small superficial lesions present as **telangiectases** (skin, lips, oral/nasal mucosa, fingers); larger lesions form visceral **AVMs** (lungs, liver, brain, GI tract, spinal cord). Lesions are fragile and prone to hemorrhage, producing recurrent epistaxis and internal bleeding (PMID: 37695357).

Key identifiers (machine-verified via EBI OLS4 and MyGene.info):

Resource	Identifier
MONDO	**MONDO:0010880** ("telangiectasia, hereditary hemorrhagic, type 2")
OMIM (phenotype)	**#600376**
OMIM (gene)	*601284 (ACVRL1)
Orphanet	ORPHA:774 (HHT, parent)
ICD-10	I78.0
MeSH	D013683
Parent term	MONDO:0019180 (hereditary hemorrhagic telangiectasia)

Related subtypes: HHT1 (MONDO:0008535, *ENG*), HHT3 (MONDO:0010996), HHT4 (MONDO:0012532), HHT5 (MONDO:0014217, *GDF2*/BMP9), and juvenile polyposis–HHT (MONDO:0008278, *SMAD4*).

Synonyms: HHT type 2; Osler–Weber–Rendu syndrome type 2; Rendu–Osler–Weber disease type 2; ACVRL1-related HHT; ALK1-related HHT.

Information source. This report is derived predominantly from **aggregated disease-level resources** (OMIM, Orphanet, MONDO, curated genotype–phenotype cohort studies, and international registries such as Cure HHT and the Danish HHT Database), supplemented by cohort-level clinical studies rather than individual EHR records.

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2. Etiology

Primary cause. HHT2 is a monogenic Mendelian disorder caused by **germline heterozygous loss-of-function mutations in *ACVRL1***. ALK1 and endoglin (ENG) are transmembrane receptors on endothelial cells that bind circulating BMP9/BMP10 and activate SMAD1/5/8; *ACVRL1* haploinsufficiency reduces this signaling and is the root cause of HHT2 (PMID: 33513792).

> "HHT is primarily associated with a reduction in endoglin (ENG) or ACVRL1 activity due to loss-of-function mutations. ENG and ACVRL1 transmembrane receptors are expressed on endothelial cells (ECs) and bind to circulating ligands BMP9 and BMP10 with high affinity. Ligand binding to the receptor complex leads to activation of the SMAD1/5/8 signalling pathway to regulate downstream gene expression." — PMID: 33513792

Genetic risk factors. The causal variant is a heterozygous *ACVRL1* mutation. Mutations cluster largely in **exons 7 and 8**, which encode the intracellular serine/threonine kinase domain (PMID: 16470787). Family history is the single strongest risk factor: almost all cases are inherited, with a 50% transmission risk to offspring. **Truncating *ACVRL1* mutations** are associated with a higher frequency of epistaxis and telangiectasis (PMID: 17224686).

Environmental / lifestyle risk factors. HHT is not caused by environmental exposures, but physiological factors modulate expression. **Blood flow / fluid shear stress** is a key mechanistic modifier—AVMs develop preferentially at sites of altered flow, and *acvrl1* transcription itself is flow- and ligand-dependent (PMID: 38727966). **Pregnancy** is a high-risk period (peripheral vasodilation, increased cardiac output) with reported heart failure, intracranial/pulmonary hemorrhage, and stroke (PMID: 28603431). Age increases lesion burden and epistaxis severity.

Protective factors. No validated genetic or environmental protective alleles are established for HHT2. Mechanistically, **restoring or enhancing ALK1 signaling** (e.g., ACVRL1 overexpression, BMP9/BMP10 supplementation) is protective in models—overexpression of *Acvrl1* prevents AVM development in both *Acvrl1*- and *Eng*-null mice (PMID: 38727966).

Gene–environment interactions. The dominant GxE axis is **flow × genotype**: physiological shear stress synergizes with BMP9/BMP10–ALK1–SMAD signaling to maintain vessel stability; ALK1 haploinsufficiency lowers this protective set point, so hemodynamic stress precipitates AVM formation (PMID: 37490341, PMID: 38727966). A "second hit" (somatic mutation, injury, inflammation, angiogenic stimulus) is thought to convert focal haploinsufficiency into a discrete lesion, explaining the sporadic, focal nature of AVMs on a uniformly heterozygous genetic background.

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3. Phenotypes

HHT2 phenotypes span symptoms, clinical signs, physical manifestations, and laboratory abnormalities. Frequencies below are drawn primarily from the French–Italian HHT network (93 HHT1, 250 HHT2), a Dutch cohort, and international QoL studies.

Phenotype	Type	HPO term	Onset	Frequency / severity in HHT2
Recurrent epistaxis	Symptom/ sign	HP:0000421 (epistaxis)	Later than HHT1; incomplete penetrance	Most common symptom; **92%** in large QoL cohort (PMID: 40055726); occurs later with incomplete penetrance (P<0.0001) (PMID: 17224686)
Mucocutaneous telangiectases	Physical sign	HP:0000562 (telangiectasia of the skin)	Progressive with age	Characteristic sites: lips, tongue, fingers, nasal mucosa
Hepatic AVM / liver involvement	Clinical sign	HP:0100761 (visceral AVM); hepatic	Adult; silent in children	**More common in HHT2**; symptomatic hepatic involvement seen only in HHT2 (PMID: 17224686); silent in 35% of HHT2 children by imaging (PMID: 28248153)
GI bleeding	Symptom	HP:0002239 (GI hemorrhage)	Adult/late-onset	**16.4%** in HHT2 vs 6.5% HHT1 (P=0.017) (PMID: 17224686)
Pulmonary AVM	Clinical sign	HP:0002110 (pulmonary AVM)	Variable	**Less frequent/smaller** than HHT1: symptomatic PAVM 5.2% vs 34.4% (P<0.001) (PMID: 17224686)
Cerebral AVM	Clinical sign	HP:0002408 (CNS AVM)	Congenital/ variable	Less common than HHT1; cerebral abscess 0.8% vs 7.5% (PMID: 17224686)
Spinal AVM	Clinical sign	HP:0002435	—	**Seen only in HHT2** (PMID: 16470787)

Phenotype	Type	HPO term	Onset	Frequency / severity in HHT2
Pulmonary arterial hypertension	Clinical sign/ lab	HP:0002092 (PAH)	Adult (occasionally infant)	~**20%** of *ACVRL1* carriers (PMID: 42516693)
Iron-deficiency anemia	Lab abnormality	HP:0001891	Secondary to bleeding	28.9% vs 15.1% in inpatients (PMID: 33779866)
Fatigue	Symptom	HP:0012378	Chronic	**79%** in QoL cohort (PMID: 40055726)

Severity and progression. Manifestations are typically **progressive** (telangiectases and epistaxis worsen with age) with **episodic** bleeding. Expressivity is highly variable even within families.

Quality-of-life impact. HHT imposes a substantial QoL burden driven chiefly by epistaxis and fatigue, with strong links to anxiety, depression, and reduced SF-36 scores. In the international Cure HHT study (n=565 completers), Epistaxis Severity Score (ESS) correlated positively with anxiety, depression, and fatigue and negatively with SF-36 (ESS vs SF-36 −26.4 [95% CI −33 to −19.9]) (PMID: 40055726).

> "The most common symptoms were epistaxis 521/565 (92%) and fatigue 446/565 (79%)." — PMID: 40055726

In the Danish HHT Database (n=124), ESS negatively correlated with SF-36 General Health (−4.14 [−6.01; −2.27], p<0.001), with age the only ESS predictor and **no association of ESS with sex, HHT type, or anticoagulation** (PMID: 42419030). Comorbid liver and GI AVMs—prominent in HHT2—further reduce physical QoL scores (PMID: 34857410).

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4. Genetic / Molecular Information

Causal gene. ***ACVRL1*** (activin A receptor-like type 1 / ALK1). HGNC:175; NCBI Gene 94; Ensembl ENSG00000139567; cytoband **12q13.13**; UniProt **P37023**; gene OMIM *601284. Protein-coding; the encoded ALK1 is a 503-amino-acid endothelial type I TGF-β/BMP receptor.

Pathogenic variants. - **Variant types:** missense (dominant in the kinase domain), nonsense, frameshift, splice-site, and small in-frame indels; large deletions/duplications also occur (detected by MLPA/del-dup analysis). - **Location:** mutations cluster largely in **exons 7 and 8**, encoding the serine/threonine kinase domain, whereas *ENG* (HHT1) mutations are widely distributed (PMID: 16470787).

> "Mutations in ACVRL1 cluster largely in exons 7 and 8, but ENG mutations were widely distributed within that gene." — PMID: 16470787

- **Classification (ACMG/AMP):** the majority of reported disease alleles are pathogenic or likely pathogenic; VUS are common for novel missense variants and can be resolved by functional assays and cosegregation.
- **Origin:** germline (heterozygous). **De novo** cases and **mosaicism** are uncommon but documented—mosaic *ENG*/*ACVRL1* mutations may be present at ≤25% allele fraction and can be missed by Sanger, requiring high-depth NGS (PMID: 21415079, PMID: 29243366).
- **Functional consequence: loss of function / haploinsufficiency** (reduced ALK1 signaling); some missense kinase-domain variants may act in a dominant-negative fashion on the receptor complex.

Allele frequency. Individually rare; genomic-database analysis (gnomAD, All of Us, Regeneron Million Exome) estimates combined *ENG+ACVRL1* pathogenic/likely-pathogenic genetic prevalence at **1.753–2.555 per 5,000**, rising to 2.874–4.327 per 5,000 including potentially pathogenic variants (PMID: 41610956).

Domain architecture (why exon 7–8 mutations abolish function; UniProt P37023, verified). ALK1 comprises: an **extracellular ligand-binding domain** (aa 22–118; cysteine-rich snake-toxin-like fold with disulfides C34–C51, C36–C41, C46–C69, C77–C89, C90–C95; N-glycosylation at N98; residues 73–76 confer BMP-ligand specificity); a single **transmembrane helix** (aa 119–141); and a cytoplasmic region (aa 142–503) containing the regulatory **GS domain** (aa 172–201; regulatory phosphoserines S155/S160/S161) and the **serine/threonine kinase domain** (aa 202–492; ATP site aa 208–216 and K229; catalytic aspartate D330). Kinase-domain (exon 7–8) mutations abolish SMAD1/5/8 phosphorylation. InterPro/Pfam: GS domain (IPR003605/PF08515), kinase domain (IPR000719/PF07714), TGFB_receptor (IPR000333). Experimental structures: extracellular domain (PDB 2LCR NMR; 4FAO, 6SF1/6SF2/6SF3, 7PPC including BMP9/BMP10 complexes) and intracellular kinase domain (PDB 3MY0, 2.65 Å, aa 195–497).

Modifier genes. The HHT phenotype is modified by other pathway members; the *ENG*/*ACVRL1*/*SMAD4*/*GDF2* network shapes severity. *SMAD4* (downstream of ALK1) loss recapitulates AVM formation and defines the JP-HHT overlap. Formal quantitative modifier loci for HHT2 severity remain incompletely defined.

Epigenetic / chromosomal information. No recurrent epigenetic signature or large-scale chromosomal abnormality is established as causal for HHT2. Large *ACVRL1* deletions are detectable by MLPA/CMA but are a minority of alleles.

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5. Environmental Information

HHT2 is a **monogenic disorder without an infectious or toxic etiology**. Environmental and lifestyle factors act only as **modifiers/triggers**:

- **Hemodynamic stress / blood flow:** the principal physiological modifier; flow regulates *acvrl1* transcription and AVMs arise at sites of abnormal shear (PMID: 38727966, PMID: 37490341).
- **Pregnancy:** raises risk of hemorrhagic and cardiovascular complications (PMID: 28603431).
- **Angiogenic stimuli / injury / inflammation:** proposed "second hits" that focalize AVM development.
- **Infectious agents:** none cause HHT2. However, pulmonary AVMs create right-to-left shunts that predispose to **paradoxical septic embolization and brain abscess**, so infection is a downstream complication—HHT is overrepresented among cerebral-abscess patients (PMID: 28578477).

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **germline heterozygous loss-of-function mutation in *ACVRL1*** (often exon 7–8, kinase domain) **results in** reduced functional ALK1 receptor on endothelial cells (haploinsufficiency).

2. Reduced ALK1 **leads to** diminished high-affinity binding of circulating **BMP9/BMP10** and **decreased phosphorylation of SMAD1/5/8** (loss of downstream transcriptional output). *(Demonstrated in vitro and in patient endothelium.)*
3. Loss of ALK1–SMAD signaling **removes the anti-angiogenic brake**: ALK1–SMAD normally synergizes with **Notch** in stalk cells to induce **HEY1/HEY2**, repressing VEGF signaling, tip-cell formation, and sprouting. Its loss **de-represses VEGF-driven angiogenesis** (PMID: 22421041). *(Demonstrated in mouse retina; rescued by BMP9.)*
4. In parallel, the **fluid-shear-stress (FSS) set point** that keeps vessels quiescent is **lowered**. Loss of SMAD4 (downstream node) **disinhibits flow-mediated KLF4–TIE2–PI3K/Akt signaling**, driving cell-cycle progression, excessive EC proliferation, and **loss of arterial identity** via KLF4 repression of CDKN2A/CDKN2B (PMID: 37490341). *(Demonstrated in* Smad4*-deletion models.)*
5. ALK1 loss **induces a KIT⁺ angiogenic endothelial-cell population** with tip-cell markers and **PI3K/KRAS activation**; **KIT is directly repressed by BMP9–ALK1–SMAD4**, so its de-repression drives AVM-like transcriptional reprogramming (PMID: 42579368). *(Demonstrated in inducible endothelial* Alk1*-knockout mice; KIT inhibition reduced malformations.)*
6. Excessive, mis-patterned angiogenesis with loss of arterial/venous identity **produces direct artery-to-vein connections** — **telangiectases** (small) and **AVMs** (large) lacking intervening capillaries.
7. These fragile, high-flow lesions **rupture and shunt**, **causing** epistaxis, GI bleeding, iron-deficiency anemia, and organ-specific sequelae (pulmonary shunt/hypoxemia/paradoxical embolism; hepatic AVM/high-output cardiac failure; cerebral/spinal AVM; PAH).

Branch point: Downstream of steps 3–5 the mechanism branches by **organ and hemodynamic context**, generating the HHT2-specific distribution (more hepatic AVM, GI bleeding, and PAH; fewer pulmonary/cerebral AVM than HHT1).

Supporting quotes:

> "ALK1-dependent SMAD signaling synergizes with activated Notch in stalk cells to induce expression of the Notch targets HEY1 and HEY2, thereby repressing VEGF signaling, tip cell formation, and endothelial sprouting." — PMID: 22421041
>
> "loss of SMAD4 disinhibits flow-mediated KLF4-TIE2-PI3K/Akt signaling, leading to cell cycle progression-mediated loss of arterial identity due to KLF4-mediated repression of cyclin dependent Kinase (CDK) inhibitors CDKN2A and CDKN2B." — PMID: 37490341

> "This process is driven by the emergence of a KIT+ angiogenic EC population with human AVM-like transcriptional features, including tip-cell markers and activation of PI3K and KRAS signaling pathways… Kit is directly repressed by BMP9-ALK1-SMAD4 signaling." — PMID: 42579368

Supporting detail

- **Molecular pathways:** BMP9/BMP10–ALK1–SMAD1/5/8 (canonical); cross-talk with **Notch–HEY1/HEY2**, **PI3K–AKT**, **KRAS/MAPK**, and **KLF4–TIE2** flow signaling. Suggested GO terms: BMP signaling pathway (GO:0030509), SMAD protein signal transduction (GO:0060395), angiogenesis (GO:0001525), regulation of sprouting angiogenesis (GO:1903670), endothelial cell proliferation (GO:0001935), arterial endothelial cell differentiation (GO:0060842), response to fluid shear stress (GO:0034405).
- **Cellular processes:** endothelial-cell proliferation, loss of arterial identity, altered tip/stalk-cell selection, and emergence of an aberrant arterial-lymphatic-like EC state that contributes to vascular leakage and AVMs (PMID: 39429196).
- **Protein dysfunction:** loss of ALK1 kinase catalytic activity (D330, K229 ATP site) → failure to transphosphorylate R-SMADs; extracellular-domain variants impair BMP9/BMP10 binding.
- **Cell types (CL terms):** endothelial cell (CL:0000115), blood vessel endothelial cell (CL:0000071), arterial endothelial cell (CL:1000413); mural cells/pericytes contribute to vessel-wall fragility.
- **Immune / tissue-damage mechanisms:** the primary injury is hemorrhagic (mechanical rupture of fragile shunts) rather than autoimmune; chronic blood loss drives iron-deficiency anemia; pulmonary shunts enable septic paradoxical embolism.
- **Molecular profiling:** single-cell/transcriptomic work identifies discrete pathogenic EC states (KIT^+ angiogenic ECs; arterial-lymphatic-like ECs) as tractable therapeutic nodes (PMID: 42579368, PMID: 39429196).

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ACVRL1 LOF (exon 7-8 kinase domain)
        │  haploinsufficiency
        ▼
 ↓ ALK1 receptor  ──►  ↓ BMP9/BMP10 binding  ──►  ↓ SMAD1/5/8-P
        │                                               │
        ▼                                               ▼
 ↓ Notch/HEY1-2 brake                       ↓ FSS set point (SMAD4)
   → de-repressed VEGF/sprouting              → KLF4-TIE2-PI3K/AKT, ↓CDKN2A/B
        │                                               │
        └────────────────►  KIT+ angiogenic EC state  ◄─┘
                            (PI3K/KRAS, tip-cell program)
                                       │
                                       ▼
                  Excess/mis-patterned angiogenesis, loss of arterial identity
                                       │
                                       ▼
                      Telangiectases + AVMs (no capillary bed)
                                       │
                 ┌──────────────┬──────┴─────┬───────────────┐
                Epistaxis     Hepatic AVM   GI bleed / anemia   PAH
                (nasal)       (HHT2 ↑)      (HHT2 ↑)           (~20%)
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7. Anatomical Structures Affected

Organ level (primary): nasal mucosa (epistaxis), skin/oral mucosa (telangiectases), **liver** (hepatic AVM—prominent in HHT2), **lungs** (pulmonary AVM), **brain** and **spinal cord** (cerebral/spinal AVM—spinal AVM seen only in HHT2), **gastrointestinal tract** (telangiectases, bleeding). **Secondary/complication organs:** heart (high-output failure from hepatic shunting; PAH), brain (paradoxical embolic stroke, abscess), bone marrow/hematologic (iron-deficiency anemia). **Body systems:** cardiovascular (primary), respiratory, digestive, nervous, integumentary.

Suggested UBERON terms: blood vessel (UBERON:0001981), endothelium of blood vessel (UBERON:0001982), nasal mucosa (UBERON:0001825), liver (UBERON:0002107), lung (UBERON:0002048), brain (UBERON:0000955), spinal cord (UBERON:0002240), gastrointestinal tract (UBERON:0001555), skin (UBERON:0002097).

Tissue and cell level: vascular endothelium (primary), with mural cell/pericyte and smooth-muscle involvement in vessel-wall remodeling. **CL terms:** endothelial cell (CL:0000115), blood vessel endothelial cell (CL:0000071), arterial endothelial cell (CL:1000413), pericyte (CL:0000669).

Subcellular level (GO Cellular Component): plasma membrane receptor complex (GO:0098802), integral component of plasma membrane (GO:0005887); intracellular signaling occurs in the cytoplasm/nucleus (SMAD nuclear translocation, GO:0005634).

Localization / lateralization: telangiectases are **multifocal and bilateral**; AVMs are typically focal and can be unilateral (e.g., a single dominant PAVM) or multiple/bilateral.

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8. Temporal Development

Onset. Insidious and age-dependent. In HHT2, **epistaxis begins later** than in HHT1 and is incompletely penetrant (PMID: 17224686). Telangiectases and visceral lesions accumulate with age; hepatic involvement is largely **silent in childhood** (detectable by imaging in ~35% of HHT2 children but clinically silent) and becomes symptomatic in adults (PMID: 28248153). PAH can rarely present in infancy (PMID: 42516693).

Progression. Chronic, lifelong, and **progressive** with **episodic** bleeding. Epistaxis severity increases with age (age is the only consistent ESS predictor) (PMID: 42419030). No formal disease "stages" exist; burden is quantified by lesion number/size and ESS.

Patterns / critical periods. Bleeding is **relapsing/episodic**; spontaneous remission of established lesions is uncommon, but antiangiogenic therapy induces telangiectasia regression. **Pregnancy** is a critical window of heightened risk (PMID: 28603431); childhood is a window for presymptomatic screening and prophylactic PAVM management.

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9. Inheritance and Population

Epidemiology. Combined HHT clinical prevalence ~**1 in 5,000** (range 1:5,000–10,000). Genetic prevalence is likely higher: **1.75–2.56 per 5,000** (pathogenic/likely-pathogenic *ENG*+*ACVRL1*), up to 2.87–4.33 per 5,000 including potentially pathogenic variants, indicating underdiagnosis

(PMID: 41610956). Regional studies: Argentina HMO 3.2/10,000 (PMID: 38683506); Alberta 1 in 3,800 with a **3.25:1 female preponderance** and elevated stroke incidence (450 vs 260/100,000; rate ratio 1.73, 95% CI 1.05–2.84) (PMID: 30520389).

> "The genetic prevalence of HHT ranged from 1.753 to 2.555 in 5000 individuals, when considering only pathogenic and likely pathogenic variants, and from 2.874 to 4.327 in 5000 individuals, when also potentially pathogenic variants were considered." — PMID: 41610956

Inheritance and genetic parameters. - **Pattern:** autosomal dominant. - **Penetrance:** age-dependent and high but incomplete—epistaxis penetrance in HHT2 is notably lower/later than HHT1 (PMID: 17224686). - **Expressivity:** highly variable, even within families. - **Genetic anticipation:** not a feature (not a repeat-expansion disorder). - **De novo / mosaicism:** rare; ~90% present with detectable heterozygous variant and nearly all have a family history; mosaicism (≤25% allele fraction) requires high-depth NGS (PMID: 29243366, PMID: 21415079). - **Founder effects/consanguinity:** *ACVRL1* mutations are highly allelic and family-specific; consanguinity is not relevant to a dominant disorder.

Population demographics. Occurs worldwide across ethnicities; specific regional founder mutations exist for HHT generally. Diagnosed cases show a **female predominance** (~3:1 in some registries), and inpatients have elevated anemia, congestive heart failure, liver disease, and cerebrovascular malformations (OR 11.04) (PMID: 33779866, PMID: 30520389).

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10. Diagnostics

Clinical criteria — Curaçao criteria (4): (1) spontaneous recurrent **epistaxis**; (2) mucocutaneous **telangiectases** at characteristic sites; (3) **visceral AVM** (pulmonary, hepatic, cerebral, spinal, GI); (4) **first-degree relative** with HHT. **≥3 = definite; 2 = possible/suspected; <2 = unlikely.** In children the criteria have low early sensitivity (only 41% met ≥3 at median 8.4 y; 63% over follow-up), so **genetic testing is important pediatrically** (PMID: 37572862).

> "In children <6 years at presentation, only 23% fulfilled at least 3 criteria initially." — PMID: 37572862

Genetic testing. Molecular confirmation by ***ACVRL1* sequencing plus deletion/duplication analysis**, typically via an **HHT gene panel** (*ENG*, *ACVRL1*, *SMAD4*, *GDF2*/BMP9). Single-gene testing is appropriate when a familial variant is known (cascade testing). WES/WGS are useful for atypical/negative cases. High-depth NGS is needed to detect **mosaicism** missed by Sanger (PMID: 29243366).

Imaging / functional screening: - **Pulmonary AVM:** contrast (agitated-saline) **transthoracic echocardiography** to detect right-to-left shunt, followed by **chest CT** (PMID: 37132738). - **Cerebral/spinal AVM: MRI**. - **Hepatic AVM: Doppler ultrasound** / CT / MRA (common-hepatic-artery dilation is an early HHT2 sign) (PMID: 28248153). - **GI bleeding:** endoscopy.

Laboratory: CBC/iron studies for chronic blood-loss anemia (HP:0001891). No single circulating biomarker is diagnostic; ESS and NOSE-HHT are validated **severity** instruments.

Differential diagnosis: *GDF2*/BMP9-related vascular-anomaly syndrome (phenotypic overlap; ~15% of clinically suspected HHT lack *ENG*/*ACVRL1*/*SMAD4* mutations) (PMID: 23972370); RASA1-related capillary malformation–AVM syndrome; CREST/systemic sclerosis; isolated PAVM; other causes of GI angiodysplasia.

Screening/cascade testing: once a familial *ACVRL1* variant is identified, **cascade genetic screening** of at-risk relatives enables presymptomatic surveillance and prophylactic PAVM embolization.

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11. Outcome / Prognosis

Survival/mortality. Life expectancy is near-normal with modern surveillance and treatment, but HHT carries excess morbidity and preventable mortality from hemorrhage, high-output cardiac failure (hepatic AVM), stroke, and brain abscess (paradoxical embolism through PAVM). HHT patients have elevated **stroke** incidence (rate ratio 1.73) (PMID: 30520389) and higher inpatient rates of congestive heart failure, liver disease, and cerebrovascular malformations (PMID: 33779866).

Morbidity/function. Chronic epistaxis, iron-deficiency anemia (frequently transfusion-dependent), and fatigue drive disability and reduced QoL; abnormal cerebral MRI findings can independently worsen depressive symptoms (BDI-II β=9.69, p=0.003) (PMID: 41950724).

Complications: paradoxical embolic stroke and **brain abscess** (PAVM), high-output heart failure and portal hypertension (hepatic AVM), PAH (~20% of *ACVRL1* carriers), hemorrhage (GI, pulmonary, intracranial), pregnancy-related events.

Prognostic factors. Higher baseline ESS, older age, extent of visceral AVM involvement, and PAH worsen prognosis. Presymptomatic **PAVM embolization** improves quality-adjusted survival (decision analysis favored immediate embolotherapy for feeding arteries ≥3 mm: 37.2 vs 32.6 QALYs) (PMID: 19376841). Genotype is prognostic for organ distribution (HHT2 → hepatic/GI/PAH) and hints at differential drug response.

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12. Treatment

Management follows the **Second International HHT Guidelines** (2020; AGREE II/GRADE methodology, 15 countries) covering epistaxis, GI bleeding, anemia, pediatric care, pregnancy, and pulmonary/hepatic/cerebral VM management (PMID: 32894695).

> "The goal of the Second International HHT Guidelines process was to develop evidence-based consensus guidelines for the management and prevention of HHT-related symptoms and complications." — PMID: 32894695

Local / interventional: - **Epistaxis:** humidification, topical therapies, laser/ablative treatment, septodermoplasty; nasal closure in refractory cases. - **Pulmonary AVM: transcatheter embolization** (coils/plugs); presymptomatic embolization recommended for feeding arteries ≥3 mm (PMID: 19376841); antibiotic prophylaxis before procedures to prevent brain abscess (PMID: 37132738). - **Hepatic AVM:** medical management first; liver transplantation for refractory high-output failure.

Systemic antiangiogenic pharmacotherapy (NCIT: antiangiogenesis agents). Three established agents target VEGF-driven angiogenesis and vessel maturation:

> "Intravenous bevacizumab, oral pazopanib, and oral thalidomide are the three targeted primary angiogenesis inhibitors, with multiple studies describing both reassuring safety and impressive effectiveness in the treatment of moderate-to-severe HHT-associated bleeding." — PMID: 35226946

Drug	Class / target	NCIT (suggested)	Evidence
Bevacizumab	Anti-VEGF monoclonal antibody	NCIT:C2039	Reduces epistaxis/GI bleeding and high-output cardiac failure (PMID: 35226946)
Pazopanib	Multikinase (VEGFR) inhibitor	NCIT:C45386	Reduces bleeding (PMID: 35226946)
Thalidomide	Antiangiogenic / vessel maturation	NCIT:C716	Bolsters vessel maturation, reduces bleeding (PMID: 28603431)
Pomalidomide	Immunomodulatory antiangiogenic	NCIT:C79809	**RCT-proven** (see below)

Pomalidomide — level-1 evidence (PATH-HHT, NCT03910244). Randomized placebo-controlled trial, 144 patients (95 pomalidomide 4 mg/day, 49 placebo, 24 weeks): mean difference in ESS change vs placebo **−0.94 points (95% CI −1.57 to −0.31)**, exceeding the 0.71-point clinically meaningful threshold; the trial was **stopped early for efficacy** (PMID: 39292928).

> "At 24 weeks, the mean difference between the pomalidomide group and the placebo group in the change from baseline in the Epistaxis Severity Score was -0.94 points (95% confidence interval [CI], -1.57 to -0.31;" — PMID: 39292928

The ATLAS extension (n=62, up to 4.4 y) confirmed durable ESS improvement (5.55→2.80, $P<0.0001$) and reduced hematologic support (9.11→5.73 red-cell/iron units, $P=0.0056$), though it was less effective for GI bleeding (PMID: 41512167).

> "Significant, durable improvements in mean epistaxis severity score (5.55 points [baseline] to 2.80 points [month 12]; P< .0001)" — PMID: 41512167

Pharmacogenomics / genotype-guided therapy. In a PATH-HHT response analysis, a **less robust pomalidomide response** was associated with older age (per 10 y, P=0.02) and **underlying *ACVRL1* (HHT2) germline mutation** (+0.62 ESS [95% CI −0.04 to 1.27] vs *ENG*/*SMAD4*/unknown, P=0.06), the first genotype signal for treatment stratification in HHT (PMID: 41719457).

> "Characteristics associated with a less robust response were older age (0.29 [95% CI, 0.05-0.53] per 10 years older, P = .02) and underlying ACVRL1 germ line mutation (0.62 [95% CI, -0.04 to 1.27] vs ENG, SMAD4, or mutation not known, P = .06]" — PMID: 41719457

Supportive care: oral/IV **iron replacement** and transfusion for anemia; antifibrinolytics (tranexamic acid) as adjuncts; antibiotic prophylaxis for PAVM.

Emerging targeted approaches (preclinical): KIT inhibition (KIT^+ angiogenic ECs), ALK1-activating therapeutics, and pathway modulators (PI3K/AKT) are under investigation (PMID: 42579368, PMID: 38727966).

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13. Prevention

Primary prevention of the genetic disease is not possible, but reproductive options (genetic counseling, **preimplantation genetic testing**, prenatal diagnosis) can prevent transmission.

Secondary prevention (early detection): cascade genetic testing of first-degree relatives once a familial *ACVRL1* variant is identified, followed by organ screening—contrast echocardiography/chest CT for PAVM, MRI for cerebral AVM, Doppler/CT for hepatic AVM. Presymptomatic **PAVM embolization** prevents stroke and brain abscess (PMID: 19376841).

Tertiary prevention: antibiotic prophylaxis before dental/surgical procedures in patients with PAVM to prevent brain abscess (PMID: 37132738); avoidance of anticoagulation where feasible; iron-status monitoring; high-risk pregnancy management (PMID: 28603431); mental-health support given the psychological impact of screening (PMID: 41950724).

Counseling. Genetic counseling is central: 50% offspring risk, variable expressivity, importance of cascade screening, and reproductive options. **Immunization/public-health/environmental** interventions are not applicable to this Mendelian disorder.

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14. Other Species / Natural Disease

Taxonomy / orthologs. *ACVRL1* is conserved across vertebrates: human *ACVRL1* (NCBI Gene 94); mouse *Acvrl1* (NCBI Gene 11482; NCBI Taxon 10090); zebrafish *acvrl1* (NCBI Taxon 7955). ALK1 signaling and its flow/BMP dependence are evolutionarily conserved—zebrafish and mouse recapitulate core AVM biology (PMID: 38727966).

Natural disease in other species. No well-characterized naturally occurring HHT2 analog is established in companion animals or wildlife (no prominent OMIA entry). HHT is essentially a human Mendelian disorder; animal involvement is via **engineered models** (Section 15) rather than spontaneous disease.

Comparative biology. The **conservation of the BMP9/BMP10–ALK1–SMAD pathway** and its flow-coupling across zebrafish and mice underpins the translational value of these models. **Zoonotic/cross-species transmission:** not applicable (non-infectious genetic disorder).

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15. Model Organisms

Mouse (Mus musculus, NCBI Taxon 10090). The workhorse model. **Inducible endothelial-specific *Alk1* (Acvrl1) knockouts** develop AVMs and reveal the KIT^{+} angiogenic EC mechanism; KIT inhibition rescued malformations (PMID: 42579368). *Acvrl1*-null and *Eng*-null mice develop AVMs preventable by *Acvrl1* overexpression (PMID: 38727966). Endothelial *Alk1* deletion also produces aberrant arterial-lymphatic-like ECs driving leakage/AVMs (PMID: 39429196); *Smad4*-deletion models define the FSS-set-point mechanism (PMID: 37490341). Available genetic tools: conditional/floxed alleles, endothelial Cre drivers, and neonatal-retina AVM assays.

Zebrafish (Danio rerio, NCBI Taxon 7955). *acvrl1* mutants and *acvrl1:egfp* reporters demonstrate blood-flow- and Bmp10-dependent *acvrl1* transcription and cranial AVM formation—ideal for studying flow–ligand coupling (PMID: 38727966). A *bmp9*-deficient zebrafish model supported the angiogenic role of BMP9 (PMID: 23972370).

Cellular / iPSC models. A **CRISPR-engineered isogenic hiPSC line carrying heterozygous *ACVRL1* c.143G>A (p.Gly48Glu)** provides a human in-vitro platform for HHT2 disease modeling and drug testing; the line retained normal karyotype, pluripotency, and trilineage differentiation (PMID: 41905104). Human endothelial cells under defined shear stress reproduce flow-dependent ALK1 regulation (PMID: 38727966).

Phenotype recapitulation and limitations. Models faithfully reproduce **AVM formation, flow-dependence, and the angiogenic EC reprogramming** central to HHT. Limitations: they may not capture the **stochastic, focal, late-onset** nature of human lesions on a heterozygous background, the full organ-distribution differences of HHT2 vs HHT1, or long-term hemorrhagic morbidity; complete knockouts are embryonic-lethal, requiring conditional/inducible strategies.

Resources: MGI (mouse *Acvrl1*), ZFIN (zebrafish *acvrl1*), IMPC/IMSR, Cellosaurus/hiPSC repositories.

Mechanistic Model / Interpretation

HHT2 is best understood as a disease of **lost endothelial angiogenic restraint**. In healthy vessels, circulating BMP9/BMP10 engage ALK1 (with endoglin) on endothelial cells and, together with physiological shear stress, activate SMAD1/5/8 to hold vessels in a quiescent, arterialized state—partly by cooperating with Notch to induce HEY1/HEY2 and repress VEGF-driven sprouting, and partly by maintaining a high fluid-shear-stress "set point" that limits proliferation. A single loss-of-function *ACVRL1* allele halves this signal. The consequences converge on **de-repressed, mis-patterned angiogenesis**: VEGF/tip-cell programs are unleashed, PI3K–AKT and KRAS pathways activate, a pathological **KIT^{+} angiogenic endothelial state** emerges, and arterial identity is lost through KLF4-mediated repression of cell-cycle inhibitors. The end result is direct artery-to-vein shunts without capillaries—telangiectases and AVMs—that are fragile and bleed.

The **HHT2-specific clinical fingerprint** (later/less-penetrant epistaxis; fewer pulmonary/cerebral AVMs; more hepatic AVMs, GI bleeding, and PAH) reflects how this common upstream lesion plays out differently across organ-specific hemodynamic and molecular contexts. This model has therapeutic corollaries realized in the clinic: **anti-VEGF and immunomodulatory antiangiogenics** (bevacizumab, pazopanib, thalidomide, pomalidomide) counteract the de-repressed angiogenesis, while emerging strategies aim upstream (ALK1-activating agents; enhancing *ACVRL1* expression) or at newly identified downstream nodes (**KIT inhibition**).

Evidence Base

PMID	Contribution
33513792	Defines LOF mechanism and BMP9/BMP10–ALK1–SMAD1/5/8 signaling
16470787	*ACVRL1* mutation clustering (exons 7–8); organ-specific AVM distribution
17224686	French–Italian genotype–phenotype cohort; HHT2 quantitative differences
42516693	ACVRL1–PAH association (~20%)
41610956	Genomic-database genetic prevalence; underdiagnosis
30520389	Population prevalence, female preponderance, elevated stroke risk
33779866	Cerebrovascular/cardiovascular comorbidity burden (NIS)
37490341	SMAD4/FSS set point; KLF4–TIE2–PI3K/Akt; loss of arterial identity
22421041	ALK1–Notch/HEY anti-angiogenic mechanism
42579368	KIT^+ angiogenic EC state; PI3K/KRAS; KIT as therapeutic target
38727966	Flow- and Bmp10-dependent *acvrl1* transcription
39429196	Arterial-lymphatic-like ECs contributing to leakage/AVM
32894695	Second International HHT Guidelines
35226946	Antiangiogenic therapies (bevacizumab, pazopanib, thalidomide)
39292928	PATH-HHT pomalidomide RCT (primary efficacy)
41512167	ATLAS long-term pomalidomide durability
41719457	ACVRL1/HHT2 trend toward weaker pomalidomide response
37572862	Low Curaçao sensitivity in children
40055726	International QoL burden (epistaxis, fatigue)
42419030	Danish ESS–QoL correlation; no difference by HHT type
34857410	Visceral AVMs reduce physical QoL
41950724	Psychological impact of cerebral MRI screening
21415079 / 29243366	Mosaicism detection

PMID	Contribution
23972370	BMP9/GDF2 differential diagnosis
19376841	PAVM embolization decision analysis
41905104	hiPSC HHT2 model (p.Gly48Glu)
28248153	Pediatric hepatic involvement in HHT2
28578477	HHT overrepresentation in cerebral abscess
28603431	Pregnancy risk and therapeutics
37132738	PAVM update; screening and antibiotic prophylaxis
16155196	Genotype–phenotype: HAVM more frequent in HHT2, distinct milder phenotype
36504622	Brain AVM cellular/molecular mechanisms review
37695357	14th HHT conference; disease overview and incidence ~1:5000

Limitations and Knowledge Gaps

- **HHT2-specific data are limited.** Much mechanistic evidence derives from pan-HHT or *Eng*/*Smad4* models; direct *Acvrl1*-heterozygous studies and HHT2-only clinical trials are fewer.
- **Modifier genes** driving the striking variable expressivity of HHT2 are largely unidentified; the "second hit" hypothesis is plausible but incompletely proven in human tissue.
- **Pomalidomide genotype effect** for *ACVRL1* is a **trend (P=0.06)**, not a definitive result, and requires prospective, genotype-stratified confirmation.
- **Penetrance and prevalence** estimates are complicated by underdiagnosis; genetic-prevalence figures assume variant pathogenicity that may be imperfect.
- **No naturally occurring animal disease** and no HHT2 newborn-screening program exist; diagnosis remains clinical/genetic and often delayed.
- **Epigenetic contributions** to HHT2 are essentially unstudied.

Proposed Follow-up Experiments / Actions

1. **Genotype-stratified antiangiogenic trials** — prospectively test whether *ACVRL1*/HHT2 patients respond less to pomalidomide/thalidomide and better to alternative agents (e.g., anti-VEGF or upstream ALK1 activators), converting the PATH-HHT signal into actionable precision medicine.
2. **KIT-directed therapy translation** — evaluate KIT inhibitors (or KIT-pathway modulators) in *Acvrl1*-mutant models and early-phase human studies, building on the KIT^{+} angiogenic EC discovery.
3. **ALK1-enhancing strategies** — test ligand supplementation (BMP9/BMP10 mimetics) and approaches that raise residual *ACVRL1* expression, exploiting the flow-dependent positive-feedback loop.
4. **HHT2 single-cell/spatial atlas** — define organ-specific endothelial states (hepatic vs pulmonary vs nasal) to explain the HHT2 clinical fingerprint and identify organ-tailored targets.
5. **Modifier-gene GWAS/exome analysis** within large HHT2 registries to explain variable expressivity and identify protective alleles.
6. **Pediatric genetic-diagnosis pathways** — formalize early molecular testing given low childhood Curaçao sensitivity, coupled with structured mental-health support for screening-detected findings.

Report compiled from 9 confirmed findings and 35 reviewed papers across 5 investigation iterations. Evidence types: predominantly human clinical/cohort and model-organism (mouse, zebrafish, hiPSC), with computational database verification of identifiers.
